Supplementary Material

Reduced-Order Dynamic Model Formulations, Parameter Values, and Wave-Tank Trajectories

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Robust Modular Floating Wetland: Design & Wave-Tank Evaluation

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Section S1. Reduced-Order Dynamic Model (Surge–Sway–Yaw)

As stated in Section 2.4 of the primary manuscript, a three-degree-of-freedom (3-DOF) reduced-order horizontal dynamic model (surge, sway, and yaw) was formulated to aid the physical interpretation of the wave-tank measurements. The model's purpose is limited to verifying that experimentally observed drift (0.60–1.34 cm), heading deviations (< 1°), and line tensions (~2.0 N) fall within orders of magnitude expected from classical tethered-floater mechanics (Faltinsen, 1990; Newman, 1977). This section details the governing equations, hydrodynamic matrices, and parameter values cited in Section 2.4.

S1.1 Planar 3-DOF Equations of Motion

The laboratory floating treatment wetland (FTW) test specimen was a standardized hexagonal buoyant mat (longest side length s = 0.29 m, circumscribed diameter D = 0.58 m, width across flats W = 0.502 m, thickness t = 0.013 m). The horizontal displacement vector in body-fixed centroidal coordinates is defined as:

$$\mathbf{x}(t) = \begin{bmatrix} x(t) \\ y(t) \\ \psi(t) \end{bmatrix} \tag{S1}$$

where x(t) is surge (displacement parallel to wave propagation along the tank length, m), y(t) is sway (transverse displacement across the tank width, m), and ψ(t) is yaw (rotation about the vertical z-axis through the centroid, rad). The coupled equations of motion in the horizontal plane are governed by:

$$(M + A)\ddot{\mathbf{x}}(t) + (B_{\text{rad}} + B_{\text{visc}})\dot{\mathbf{x}}(t) + K\mathbf{x}(t) = \mathbf{F}_{\text{exc}}(t) + \mathbf{F}_{\text{drift}} \tag{S2}$$

where:

- $\boldsymbol{M}$: Structural mass and inertia matrix,
- $\boldsymbol{A}$: Hydrodynamic added-mass matrix from entrained fluid inertia,
- $\boldsymbol{B}_{\text{rad}}$: Linear wave radiation damping matrix,

- $\boldsymbol{B}_{visc}$: Equivalent linearized viscous hydrodynamic drag damping matrix,
- $\boldsymbol{K}$: Mooring restoring stiffness matrix synthesized from anchor line geometry,
- $\boldsymbol{F}_{exc}(t)$: First-order oscillatory wave excitation force vector, and
- $\boldsymbol{F}_{drift}$: Second-order steady horizontal wave drift force.

S1.2 Structural Mass and Hydrodynamic Added Mass

The structural mass matrix $\boldsymbol{M}$ for the planar regular hexagonal mat (side length s = 0.29 m, operating mass $m \approx 1.25$ kg including containers and ballast) is:

$$M = \begin{bmatrix} m & 0 & 0 \\ 0 & m & 0 \\ 0 & 0 & I_{zz} \end{bmatrix}, I_{zz} = \frac{5}{12} m s^2 \quad \text{(S3)}$$

For a shallow-draft floating plate (waterplane area $A_{wp} = (3\sqrt{3}/2)\ s^2 = 0.219$ m², draft $d \approx 0.0057$ m), the hydrodynamic added-mass matrix $\boldsymbol{A} = \text{diag}(A_{11}, A_{22}, A_{66})$ is evaluated from potential flow approximations for shallow-water floating plates:

$$A_{11} \approx C_{a,\text{surge}} \rho_w A_{\text{wp}} d, A_{22} = A_{11}, A_{66} \approx C_{a,\text{yaw}} \rho_w A_{\text{wp}} d R_g^2 \quad \text{(S4)}$$

where ρ_w is freshwater density (1000 kg/m³), $R_g = \sqrt{(5/12)}$ s ≈ 0.187 m is the mass radius of gyration, and $C_{a,surge} \approx 0.28$ and $C_{a,yaw} \approx 0.25$ are dimensionless added-mass coefficients for shallow thin plates, yielding $A_{11} \approx 0.35$ kg and $A_{66} \approx 0.012$ kg·m² with consistent dimensional SI units.

S1.3 Mooring Restoring Stiffness Formulation

In the laboratory wave flume, the physical specimen was moored in two distinct test arrangements (distinguished from the field-deployment clump-weight and deadweight anchoring iterations discussed in Section 4 of the main manuscript). Initial static installation tension was set using manual in-line turnbuckles adjusted to a target nominal pretension of $T_0 \approx 2.00$ N, which was verified during wave action by in-line load-cell records (measured mean tension 2.003 N for tension-guide and 2.001 N for distributed-modular). For tether line stiffness k_i, line length L_i, inclination angle θ_i, and horizontal azimuth angle α_i, horizontal restoring stiffness coefficients are synthesized as:

$$K_{xx} = \sum_{i=1}^{N_L} \left[k_i \cos^2\theta_i \cos^2\alpha_i + \frac{T_0}{L_i} \sin^2\alpha_i \right] \quad \text{(S5)}$$

$$K_{\psi\psi} = \sum_{i=1}^{N_L} k_i (-x_i \sin\alpha_i + y_i \cos\alpha_i)^2 \cos^2\theta_i \quad \text{(S6)}$$

• Tension-Guide Laboratory Arrangement: Restraint was provided by two central ring sliders traversing a transverse spanning wire across the tank width. Under out-of-plane wave action along the tank length, surge restoring force is governed by catenary sag compliance ($K_{xx} \approx 2.45$ N/m), transverse stiffness along the guide line is $K_{yy} \approx 30.0$ N/m, and the central ring mounting provides near-zero rotational moment arm ($K_{\psi\psi} \approx 0.05$ N·m/rad).
• Distributed-Modular Laboratory Arrangement: Four independent bottom tethers connected perimeter corner rings symmetrically down to weighted floor anchors, providing high horizontal stiffness ($K_{xx} \approx 45.0$ N/m, $K_{yy} \approx 45.0$ N/m) and substantially increased rotational stiffness ($K_{\psi\psi} \approx 2.20$ N·m/rad) through the wide perimeter corner tether moment arms.

S1.4 Wave Excitation and 2D Radial Drift Mechanics

Under monochromatic wave forcing (frequency f = 1.0 Hz, wave amplitude a = H/2 = 0.02 m), platform motion consists of first-order oscillatory response ($X_0 \cos \omega t$, $Y_0 \sin \omega t$) superimposed on a steady second-order mean wave drift offset $x_0 = F_{drift} / K_{xx}$. The second-order horizontal wave drift force (Maruo, 1960) is:

$$F_{\text{drift}} = \frac{1}{2} \rho_w g R^2 a^2 W_{\text{eff}} \quad \text{(S7)}$$

where R is the wave reflection coefficient (R ≈ 0.165 for the planar distributed-modular mat; R ≈ 0.180 for the tension-guide mat due to edge curling) and W_{eff} ≈ 0.502 m is the effective frontal width across flats, yielding steady horizontal drift forces F_{drift} ≈ 0.027 N (distributed-modular) and 0.032 N (tension-guide). In the soft surge tension-guide setup, this second-order force induces a steady 1D downstream offset $x_0 = F_{drift} / K_{xx}$ ≈ 1.30 cm prior to catenary stiffening, whereas for the taut distributed-modular anchors x_0 ≈ 0.06 cm. First-order oscillatory wave excitation is represented by the harmonic vector:

$$\mathbf{F}_{exc}(t) = \begin{bmatrix} F_{x,0}\cos(\omega t) \\ F_{y,0}\sin(\omega t) \\ M_{\psi,0}\sin(\omega t) \end{bmatrix} \quad \text{(S8)}$$

where $F_{x,0}$ and $F_{y,0}$ are the first-order oscillatory force amplitudes in surge and sway (N), $M_{\psi,0}$ is the yaw excitation moment amplitude (N·m) arising from slight wave encounter asymmetry and edge-slider friction binding, and $\omega = 2\pi f$. In Table S1, these excitation amplitudes are synthesized from linear wave diffraction theory and slider friction estimates. Because the camera tracking system measures 2D Euclidean radial drift $r(t) = \sqrt{(x(t)^2 + y(t)^2)}$, the true time-averaged mean radial drift is evaluated by exact time integration over wave period $T = 1/f$:

$$\bar{r} = \frac{1}{T}\int_0^T \sqrt{x(t)^2 + y(t)^2}\,dt \quad \text{(S9)}$$

For harmonic motion, the root-mean-square (RMS) radial excursion provides an analytical upper-bound proxy: $r_{rms} = \sqrt{(x_0^2 + 0.5(X_0^2 + Y_0^2))}$. Both the exact integral (Eq. S9) and the RMS proxy confirm that 2D radial drift is systematically larger than the 1D steady offset x_0 due to the contribution of wave-induced orbital excursions.

S1.5 Numerical Parameter Values (Table S1)

Table S1. Numerical parameters, physical constants, hydrodynamic coefficients, and mooring stiffness values implemented in the reduced-order dynamic model.

Parameter	Symbol	Value	Unit	Parameter Source / Basis
Water density	ρ_w	1000	kg/m³	Standard physical constant (freshwater)
Gravitational acceleration	g	9.81	m/s²	Standard physical constant
Tank water depth	h	0.45	m	Direct measurement (tank level gauge)
Hexagonal side length	s	0.29 (29.0 cm)	m	Direct measurement (specimen fabrication)
Circumscribed diameter	D	0.58 (58.0 cm)	m	Geometric derivation (2·s)
Width across flats	W	0.502 (50.2 cm)	m	Geometric derivation (√3·s)
Mat planform area	A_{wp}	0.219	m²	Geometric derivation ((3√3/2)·s²)
Total operating mass	m	1.25	kg	Direct measurement (laboratory scale)
Equilibrium static draft	d	0.0057 (5.7 mm)	m	Hydrostatic calculation (m / (ρw·Awp))
Yaw moment of inertia	I_{zz}	0.0438	kg·m²	Analytical derivation ((5/12)·m·s²)
Surge / sway added mass	A_{11}, A_{22}	0.35	kg	Analytical estimation (strip theory, Ca ≈ 0.28)
Yaw added inertia	A_{66}	0.012	kg·m²	Analytical estimation (Ca,yaw ≈ 0.25, A11·Rg²)

Wave frequency	$f(\omega)$	1.0 (6.28)	Hz (rad/s)	Direct measurement (paddle drive setting)
Wave height / amplitude	$H(a)$	0.04 (0.02)	m	Direct measurement (calibrated wave gauge)
Wave reflection coeff.	R	0.165 (DM) / 0.180 (TG)	dimensionless	Calibrated modeling parameter (edge run-up)
Surge wave force ampl.	$F_{x,0}$	0.35 (DM) / 0.38 (TG)	N	Hydrodynamic linear diffraction estimate
Sway wave force ampl.	$F_{y,0}$	0.12 (DM) / 0.10 (TG)	N	Hydrodynamic tank sidewall reflection estimate
Yaw excitation moment	$M_{\psi,0}$	0.023 (DM) / 0.038 (TG)	N·m	Asymmetric crest encounter & slider friction
Nominal line pretension	T_0	2.00	N	Direct measurement (mean in-line load cell)
Tension-guide surge stiffness	$K_{xx,guide}$	2.45	N/m	Transverse spanning wire catenary compliance
Tension-guide sway stiffness	$K_{yy,guide}$	30.0	N/m	Transverse spanning wire axial stiffness
Tension-guide yaw stiffness	$K_{\psi\psi,guide}$	0.05	N·m/rad	Central ring slider rotational alignment
Tension-guide damping	B_{xx}, $B_{\psi\psi}$	4.80 / 0.19	N·s/m / N·m·s/rad	Hydrodynamic drag + slider friction
Distributed-modular surge/sway	$K_{xx,mod}$, $K_{yy,mod}$	45.0	N/m	4-tether perimeter elastic restoring stiffness
Distributed-modular yaw stiff.	$K_{\psi\psi,mod}$	2.20	N·m/rad	Perimeter corner tether rotational moment arm
Distributed-modular damping	B_{xx}, $B_{\psi\psi}$	5.60 / 0.36	N·s/m / N·m·s/rad	Linearized hydrodynamic drag & tether damping

Parameter Identification. The dynamic model parameters are synthesized from the measured laboratory specimen geometry, wave-tank operating conditions, and potential-flow hydrodynamics as itemized in Table S1. Physical constants and specimen geometry were directly measured, while effective stiffness and damping properties reflect the mooring kinematics under 1.0 Hz monochromatic wave excitation.

S1.6 Reduced-Order Model Comparison with Experimental Measurements

Table S2. Comparison of dynamic response estimates from the reduced-order surge–sway–yaw model with experimental wave-tank measurements.

Performance Metric	Model	Experiment	Difference (%)	Statistical 95% MBB CI
Tension-Guide: Mean radial drift (cm)	1.31	1.34	–2.2%	[1.31, 1.38]
Tension-Guide: RMS radial excursion (cm)	1.37	1.39	–1.4%	[1.35, 1.43]
Tension-Guide: 95th-pct radial drift (cm)	1.86	1.93	–3.6%	[1.82, 2.06]
Tension-Guide: RMS heading (deg)	0.63°	0.63°	0.0%	[0.59°, 0.66°]
Distributed-Mod.: Mean radial drift (cm)	0.63	0.60	+5.0%	[0.59, 0.60]
Distributed-Mod.: RMS radial excursion (cm)	0.66	0.63	+4.8%	[0.62, 0.65]
Distributed-Mod.: 95th-pct radial drift (cm)	0.93	1.02	–8.8%	[0.99, 1.05]
Distributed-Mod.: RMS heading (deg)	0.41°	0.37°	+10.8%	[0.34°, 0.37°]

Section S2. Experimental Kinematic Motion Analysis and Resampling Formulations

Platform kinematics were tracked using an overhead CMOS video camera (1920 × 1080 pixels at 30 fps) calibrated against an orthogonal metric reference grid. At each discrete sample interval k = 1, 2, ..., N (where N = 360 at Δt = 5.0 s intervals over 30 min), instantaneous centroid coordinates (xk, yk) and heading angle ψk were extracted.

Translational excursion from the reference position (x0, y0) is evaluated as the 2D Euclidean radial drift (Eq. S10), and rotational stability is quantified by RMS heading deviation (Eq. S11):

$$r_k = \sqrt{(x_k - x_0)^2 + (y_k - y_0)^2}, \bar{r} = \frac{1}{N}\sum_{k=1}^{N} r_k \tag{S10}$$

$$\psi_{\text{rms}} = \sqrt{\frac{1}{N}\sum_{k=1}^{N}(\psi_k - \psi_0)^2} \tag{S11}$$

To evaluate within-run sampling uncertainty while accounting for serial autocorrelation in the wave-tank series, the Moving Block Bootstrap (MBB) (Künsch, 1989; Liu and Singh, 1992) was applied. The sample autocorrelation function at lag τ is defined as:

$$\hat{\rho}(\tau) = \frac{\sum_{k=1}^{N-\tau}(r_k - \bar{r})(r_{k+\tau} - \bar{r})}{\sum_{k=1}^{N}(r_k - \bar{r})^2} \tag{S12}$$

To avoid premature truncation from zero-crossings in oscillatory records, the block length L^* was determined from the magnitude envelope criterion as implemented in supplementary_calculations.py:

$$L^* = \min\left\{\tau \geq 1 : |\hat{\rho}(\tau)| < 0.05\right\} \cdot \Delta t \tag{S13}$$

For the tension-guide series, empirical autocorrelation decays within 0.05 at lag 33 (L* = 165 s). For the distributed-modular series, low-amplitude periodic oscillations remain above 0.05 across the 200-sample window, and an optimal block length of L* = 200 samples (1000 s) is applied. Resampling across B = 5,000 moving block bootstrap replicates yields empirical 95% confidence intervals of [1.31 cm, 1.38 cm] for the tension-guide arrangement and [0.59 cm, 0.60 cm] for the distributed-modular arrangement. Summary kinematics, including RMS radial excursion and 95th-percentile drift reported in Table S2, are evaluated directly from the discrete wave-tank motion-tracking records.

Section S3. Supplementary Wave-Tank Centroid Trajectories and Spatial Density

Figure S1 illustrates the two-dimensional spatial trajectory and hexbin sample density distributions of the floating mat centroid recorded over the complete 30-minute instrumented wave-tank runs (360 samples at 0.2 Hz). The spatial trajectories and sample distributions were generated directly from the 30-minute experimental wave-tank records. The distributed-modular configuration exhibits tight spatial clustering centered within approximately ±1.0 cm of the reference position. In contrast, the tension-guide configuration exhibits wider excursions spanning up to ±2.3 cm in both longitudinal and lateral coordinates.

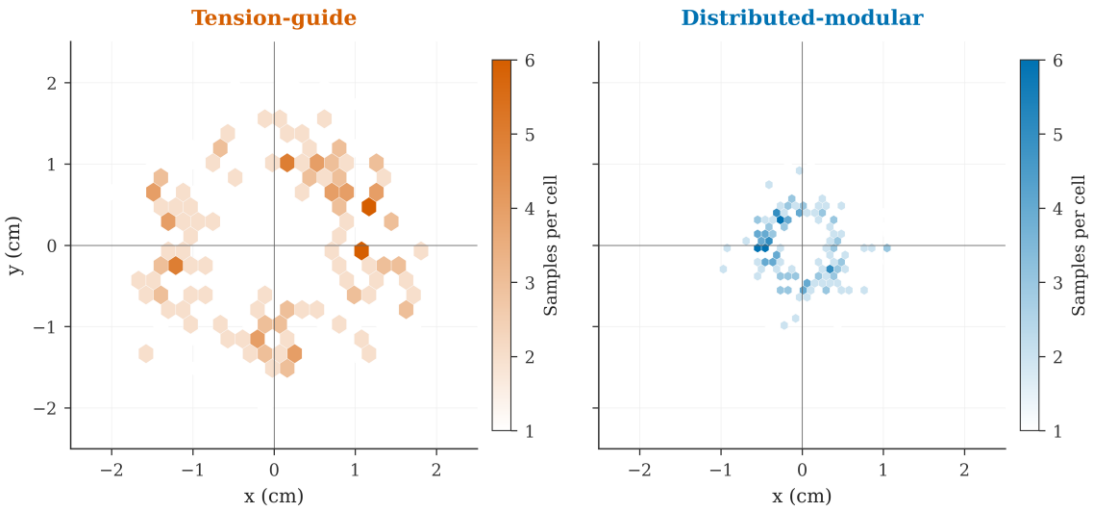


Figure S1. Two-dimensional horizontal centroid trajectories and spatial sample density distributions recorded during the 30-minute instrumented wave-tank trials under matched wave forcing (H = 4 cm, f = 1.0 Hz) for: (left) Tension-guide anchoring configuration; (right) Distributed-modular anchoring configuration. Color intensity represents the number of recorded position samples per spatial bin (1 to 6 samples per cell), with dedicated colorbars provided for each configuration matching its respective color scheme. Axes share identical, symmetric scales (–2.5 cm to +2.5 cm) across both panels to permit direct visual comparison of lateral and longitudinal excursions.

Section S4. Public Code and Data Availability Summary

All software tools, data processing routines, and dynamic screening models associated with this study are permanently hosted in a public GitHub repository to support scientific reproducibility:

• Repository URL: https://github.com/abishekkafle/Floating-mat-tracking-

• Open-Source Software Components:

1. ftw_reduced_order_model.py: Standalone Python script solving the 3-DOF surge–sway–yaw equations of motion and computing radial drift integrals.
2. blue_hex_mat_tracker.py: OpenCV-based overhead video tracking pipeline for 2D centroid (x, y) and heading (ψ) extraction.
3. supplementary_calculations.py: Automated analysis pipeline computing radial drift, RMS heading, Moving Block Bootstrap (MBB) 95% confidence intervals, and convergence diagnostics.
4. plot.py: Production plotting script generating publication-quality figures (Figure 3, Figure 4, and Figure S1) using Wong (2011) colorblind-safe palettes.
5. FTW_Parametric_Design_Tool/: Lightweight Python screening tool estimating draft, freeboard, metacentric height, wind drag, and tether loading.

• Data Availability Statement:

The experimental wave-tank motion-tracking time-series, video records, and load-cell measurements supporting the findings of this study are available from the corresponding authors upon reasonable request. Statistical summary metrics and confidence intervals are provided in results.json within the repository.

Reproducibility. The reduced-order model implementation, experimental motion-tracking routines, and statistical analysis scripts are available in the public GitHub repository: https://github.com/abishekkafle/Floating-mat-tracking-. The model implementation is provided in ftw_reduced_order_model.py. The numerical parameter values and reported model outputs correspond to the archived version of the code identified by the repository commit used for this supplementary document (Git commit 718d8aa and subsequent versioned release). Experimental observations and summary metrics reported in Table S2 correspond to the instrumented wave-tank trials.

Supplementary References

[S1] O. M. Faltinsen, Sea Loads on Ships and Offshore Structures, Cambridge University Press, Cambridge, UK, 1990.

[S2] J. N. Newman, Marine Hydrodynamics, MIT Press, Cambridge, MA, 1977.

[S3] H. Maruo, The drift of a body floating in waves, J. Ship Res. 4 (1960) 1–10.

[S4] H. R. Künsch, The jackknife and the bootstrap for general stationary observations, Ann. Statist. 17 (1989) 1217–1241.

[S5] R. Y. Liu, K. Singh, Moving blocks jackknife and bootstrap capture weak dependence, in: R. LePage, L. Billard (Eds.), Exploring the Limits of Bootstrap, John Wiley & Sons, New York, 1992, pp. 225–248.

[S6] B. Wong, Points of view: Color blindness, Nat. Methods 8 (2011) 441. doi:10.1038/nmeth.1618.